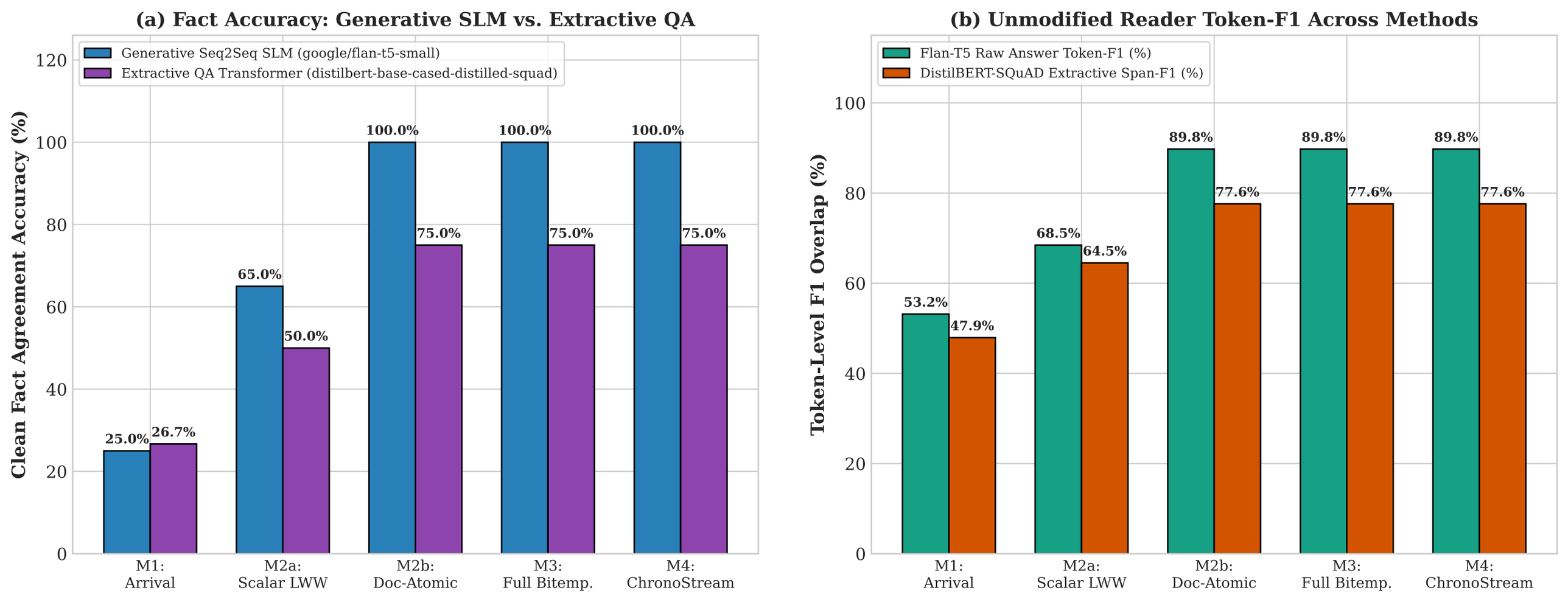
(a) Fact Accuracy: Generative SLM vs. Extractive QA
Generative Seq2Seq SLM (google/flan-t5-small)
Extractive QA Transformer (distilbert-base-cased-distilled-squad)
Clean Fact Agreement Accuracy (%)
0
20
40
60
80
100
120
25.0%
26.7%
65.0%
50.0%
100.0%
75.0%
100.0%
75.0%
100.0%
75.0%
M1: Arrival
M2a: Scalar LWW
M2b: Doc-Atomic
M3: Full Bitemp.
M4: ChronoStream
(b) Unmodified Reader Token-F1 Across Methods
Flan-T5 Raw Answer Token-F1 (%)
DistilBERT-SQuAD Extractive Span-F1 (%)
Token-Level F1 Overlap (%)
0
20
40
60
80
100
53.2%
47.9%
68.5%
64.5%
89.8%
77.6%
89.8%
77.6%
89.8%
77.6%
M1: Arrival
M2a: Scalar LWW
M2b: Doc-Atomic
M3: Full Bitemp.
M4: ChronoStream